

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper: BCS-403 — Fundamentals Of Computing

Time: Three hours

Maximum Marks: 70

*Instructions: Answer any five questions, including question no. 1, which is compulsory.***Question 1.** Write the full name and function of the following terms: [2 × 7 = 14]

- (a) UNIVAC I
- (b) ULSI
- (c) CD-ROM
- (d) ARPANET
- (e) FTP
- (f) URL
- (g) HTTP

Question 2.

- (a) What is computer and what are its characteristics? Explain each. [7]
- (b) Differentiate among computer program, software and hardware. Explain the system software and application software with appropriate examples. [2+5]

Question 3.

- (a) Divide $(1010110)_2$ by $(110)_2$. [3]
- (b) Multiply $(10101)_2$ with $(101)_2$. [2]
- (c) Add these binary numbers: 11111, 11111, 10111, 1011, 10111. [3]
- (d) Convert the binary number 11011101.0111_2 to decimal number system. [3]
- (e) Convert the decimal number 352_{10} to binary number system. [3]

Question 4.

- (a) What do you mean by software development steps? Explain each step. [7]
- (b) What is programming language? Describe the different types of programming language with merits and demerits of each. [7]

Question 5.

- (a) What is Internet? How is it different from the World Wide Web? Write the various applications of Internet. [2+2+3]
- (b) What are the steps involved in solving problems using computers? Explain each. [7]

Question 6.

- (a) Write the characteristics of an algorithm. Draw a flowchart to check if a given year is a leap year or not. [3+4]
- (b) What is pseudocode? Write a pseudocode to find the sum of digits of a given number. [2+5]

Question 7. Write short notes on the following:

- (a) Evolution of Computers [4]
- (b) Structured Programming and Logic/Control Structures [5]
- (c) Flowchart with its merits and demerits. [5]